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ON

**SMART COMPLAINT MANAGEMENT
SYSTEM**

COMPUTER SCIENCE AND ENGINEERING
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ABSTRACT

The Smart Complaint Management System is a comprehensive web-based application developed to modernize and simplify complaint handling and administrative management in residential societies, educational institutions, offices, and organizational environments. Traditional complaint handling methods rely heavily on paper registers, phone calls, emails, or informal messaging applications, which often lead to communication gaps, delays in issue resolution, loss of records, and lack of transparency. The proposed system addresses these challenges through a centralized digital platform that allows residents and users to raise complaints, track issue progress, and receive updates in real time.

The application provides dedicated modules for complaint registration, complaint tracking, staff management, role assignment, salary management, and announcement broadcasting. Administrators can monitor all complaints through a dashboard, assign tasks to staff members, update issue status, and maintain organized digital records for future reference. The system ensures better coordination between residents and management while reducing manual work and increasing accountability.

The project is developed using modern web technologies including HTML, CSS, JavaScript, backend frameworks, and secure database integration. The user-friendly interface, structured workflow, and secure authentication mechanisms make the platform reliable and scalable. The expected outcome of the project is an efficient, transparent, and user-centric complaint management solution capable of improving communication, operational efficiency, and service quality within residential and institutional environments.

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1. INTRODUCTION

1.1 Background

In today's fast-growing digital world, efficient communication and problem-solving systems have become essential for residential societies, educational institutions, and organizations. Traditional complaint handling methods often involve manual registers, emails, phone calls, or verbal communication, which are inefficient and difficult to track. Such methods usually lead to delays, lack of accountability, and poor transparency between residents and management.

The Smart Complaint Management System is designed to eliminate these limitations by providing a centralized web-based platform where users can register complaints and monitor their status in real time. The system digitizes complaint handling and simplifies society management operations by integrating multiple administrative functionalities into a single platform.

1.2 Problem Statement

Many residential societies and organizations still depend on outdated complaint management processes. These systems face several challenges such as:

- Delays in complaint registration and issue resolution.
- Poor record maintenance and tracking mechanisms.
- Lack of transparency between users and administrators.
- Difficulty in assigning complaints to responsible staff members.
- Manual management of staff salaries and announcements.
- Absence of centralized communication systems.

The absence of a digital management system creates inefficiency, confusion, and

communication

gaps among residents, staff, and administrators.

1.3 Objectives

The primary objectives of the Smart Complaint Management System are:

- To provide a secure online platform for complaint registration.
- To allow residents to track complaint status in real time.
- To enable administrators to assign and monitor complaints efficiently.
- To maintain digital records of complaints and administrative activities.
- To simplify staff and salary management processes.
- To improve transparency and communication between users and management.

2. SOFTWARE AND HARDWARE REQUIREMENTS

2.1 Development Methodology

The project follows a structured software development methodology that includes requirement analysis, system design, implementation, testing, and deployment. Agile development practices

were followed to ensure continuous improvement and timely feature integration.

The development process involved:

- Requirement gathering and feasibility analysis.
- Designing frontend and backend architecture.
- Database schema creation.
- Module-wise implementation and testing.
- Performance optimization and deployment.

2.2 Programming Environment

Frontend Development:

- HTML for webpage structure.
- CSS for responsive design and styling.
- JavaScript for interactivity and client-side logic.

Backend Development:

- Node.js and Express.js for server-side functionality.
- REST APIs for frontend-backend communication.

Database:

- MySQL/PostgreSQL database for storing complaint records, user data, and staff details.

Development Tools:

- Visual Studio Code.
- Git and GitHub for version control.

2.3 Software Requirements

- Operating System: Windows / Linux / macOS
- Browser Support: Google Chrome, Mozilla Firefox, Edge
- Node.js Runtime Environment
- Database Server
- Internet Connection for deployment and synchronization

2.4 Hardware Requirements

- Processor: Intel Core i5 or equivalent
- RAM: Minimum 8 GB
- Storage: Minimum 1 GB free space
- Stable Internet Connection
- Display Resolution: 1366x768 or higher

3. DATABASE ANALYSIS, DESIGN AND IMPLEMENTATION

3.1 Database Overview

The Smart Complaint Management System uses a relational database structure to store and manage complaint information, resident records, announcements, staff details, salary information, and role assignments. The database ensures secure storage and efficient retrieval of data while maintaining data integrity and consistency.

The database is designed to support multiple users simultaneously while ensuring smooth system performance and scalability.

3.2 Database Entities

The system contains the following major entities:

- User: Stores resident and administrator information.
- Complaint: Maintains complaint details and status.
- Staff: Stores employee details and assigned responsibilities.
- Announcement: Contains notices and updates sent by administrators.
- Salary: Maintains salary records of staff members.
- Role Management: Defines user permissions and responsibilities.

3.3 Relationships

- One user can raise multiple complaints.
- One administrator can manage many complaints.
- One staff member can handle multiple assigned tasks.
- Announcements are visible to all registered residents.

- Salary records are linked to individual staff members.

3.4 Database Advantages

The database structure ensures:

- Fast retrieval of complaint data.
- Secure authentication and authorization.
- Proper maintenance of historical records.
- Efficient handling of multiple users simultaneously.
- Better reporting and analytics capabilities.

4. PROGRAM STRUCTURE ANALYSIS AND GUI CONSTRUCTION

4.1 Home Page

The homepage provides quick access to login, complaint registration, announcements, and important notices.

The module is designed with a user-friendly interface to ensure smooth navigation and ease of use. Responsive layouts and intuitive controls improve accessibility and user experience across different devices and platforms.

4.2 Login and Registration

Users can securely register and log into the system with role-based access control.

The module is designed with a user-friendly interface to ensure smooth navigation and ease of use. Responsive layouts and intuitive controls improve accessibility and user experience across different devices and platforms.

4.3 Complaint Registration

Residents can submit complaints related to maintenance, plumbing, electricity, and security.

The module is designed with a user-friendly interface to ensure smooth navigation and ease of use. Responsive layouts and intuitive controls improve accessibility and user experience across different devices and platforms.

4.4 Complaint Tracking

Users can monitor complaint progress and resolution updates in real time.

The module is designed with a user-friendly interface to ensure smooth navigation and ease of use. Responsive layouts and intuitive controls improve accessibility and user experience across different devices and platforms.

4.5 Admin Dashboard

The admin dashboard displays all complaints, statistics, and management tools in a centralized interface.

The module is designed with a user-friendly interface to ensure smooth navigation and ease of use. Responsive layouts and intuitive controls improve accessibility and user experience across different devices and platforms.

4.6 Staff Management

Administrators can add, update, and manage staff information and assigned tasks.

The module is designed with a user-friendly interface to ensure smooth navigation and ease of use. Responsive layouts and intuitive controls improve accessibility and user experience across different devices and platforms.

4.7 Announcement Module

Important announcements and notices can be broadcasted to all users instantly.

The module is designed with a user-friendly interface to ensure smooth navigation and ease of use. Responsive layouts and intuitive controls improve accessibility and user experience across different devices and platforms.

4.8 Salary Management

The system maintains salary records and payment details for staff members.

The module is designed with a user-friendly interface to ensure smooth navigation and ease of use. Responsive layouts and intuitive controls improve accessibility and user experience across different devices and platforms.

4.9 Role Management

Different roles such as admin, staff, and residents are managed with proper access permissions.

The module is designed with a user-friendly interface to ensure smooth navigation and ease of use. Responsive layouts and intuitive controls improve accessibility and user experience across different devices and platforms.

4.10 User Profile

Users can update personal details and manage account information securely.

The module is designed with a user-friendly interface to ensure smooth navigation and ease of use. Responsive layouts and intuitive controls improve accessibility and user experience across different devices and platforms.

5. IMPLEMENTATION AND MODULES

Authentication Module

Implements secure login and session management using encrypted credentials and role-based authorization.

The implementation process involved frontend-backend integration, database connectivity, API testing, and performance optimization. The modular structure improves maintainability and future scalability of the system.

Complaint Module

Handles complaint submission, categorization, assignment, and status updates.

The implementation process involved frontend-backend integration, database connectivity, API testing, and performance optimization. The modular structure improves maintainability and future scalability of the system.

Admin Control Panel

Provides centralized management tools for administrators.

The implementation process involved frontend-backend integration, database connectivity, API testing, and performance optimization. The modular structure improves maintainability and future scalability of the system.

Notification System

Ensures users receive timely updates and announcements.

The implementation process involved frontend-backend integration, database connectivity, API testing, and performance optimization. The modular structure improves maintainability and future scalability of the system.

Database Connectivity

Integrates backend APIs with the database for secure data storage and retrieval.

The implementation process involved frontend-backend integration, database connectivity, API testing, and performance optimization. The modular structure improves maintainability and future scalability of the system.



6. PROJECT SNAPSHOTS

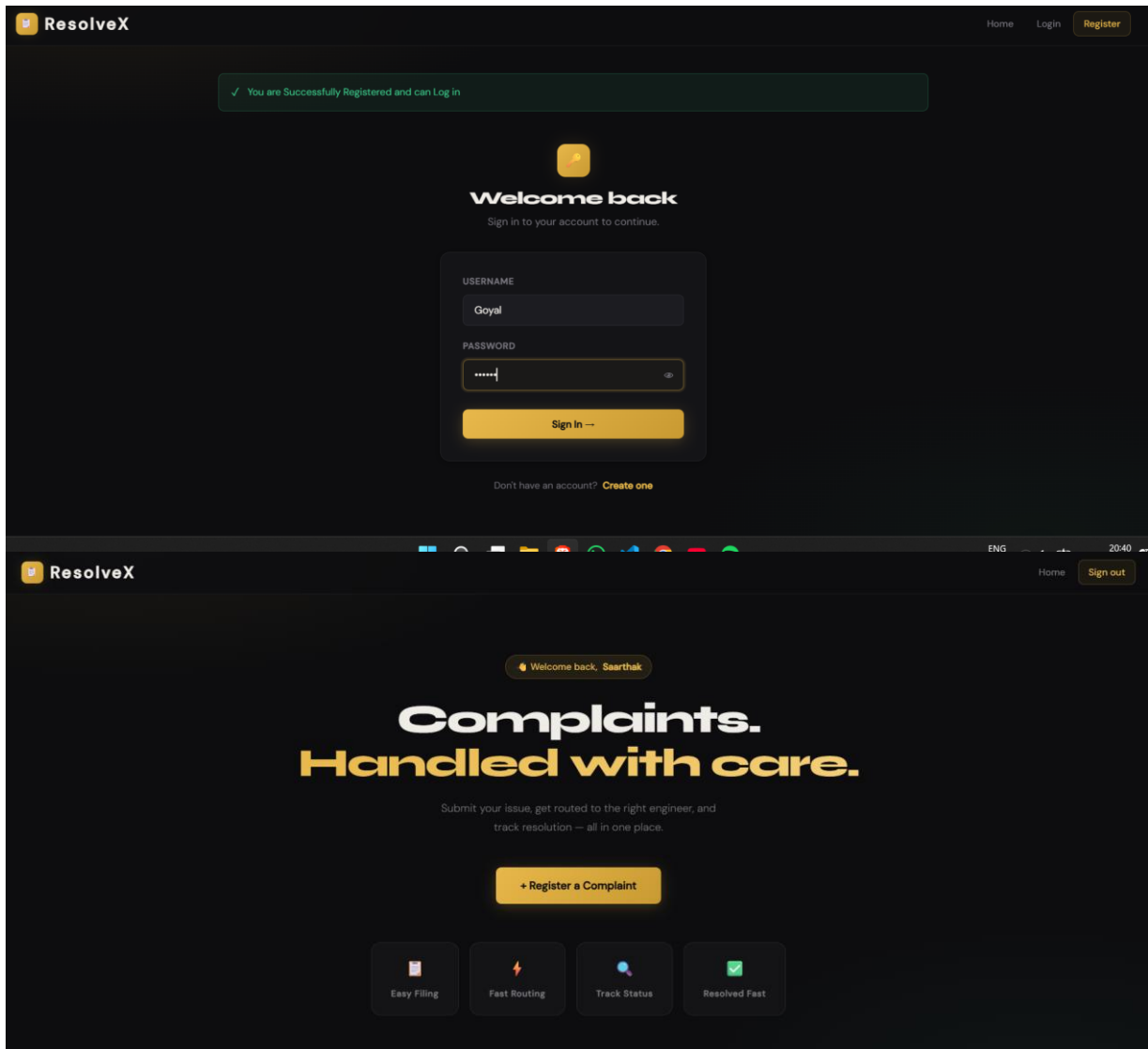
The image displays two screenshots of a web application named "ResolveX".

Top Screenshot (Login Page):

- URL: `http://localhost:3000/login`
- Header: ResolveX, Home, Login, Register
- Message: "Welcome back" with a subtext "Sign in to your account to continue."
- Form fields: USERNAME (placeholder: "Enter your username"), PASSWORD (placeholder: "Enter your password", with an eye icon for visibility toggle).
- Button: "Sign In →"
- Link: "Don't have an account? [Create one](#)"

Bottom Screenshot (Registration Page):

- Header: ResolveX, Home, Login, Register
- Message: "Create account" with a subtext "Join the complaint management system."
- Form fields: FULL NAME (Saarthak), USERNAME (Goyal), EMAIL ADDRESS (saarthak756.be22@chitkara.edu.in), PASSWORD (masked with asterisks), CONFIRM (masked with asterisks), ROLE (User).
- Role description: "User — Can register and track complaints."
- Button: "Create Account →"





← Back

Register a Complaint

Fill in the details below. We'll connect you to the right engineer.

FULL NAME

Saarthak Goyal

EMAIL ADDRESS

saarthak756.be22@chitkara

CONTACT NUMBER

+91 95014 68460

COMPLAINT DESCRIPTION

I have lost my laptop's charger and i have doubt someone stole it .

Be as specific as possible for faster resolution.

Submit Complaint →

EMAIL ADDRESS

saarthak756.be22@chitkara

CONTACT NUMBER

+91 95014 68460

COMPLAINT DESCRIPTION

I have lost my laptop's charger and i have doubt someone stole it .

Be as specific as possible for faster resolution.

Submit Complaint →



~24 hrs

Average response



Secure

End-to-end tracked



Expert

Assigned engineers

```
✓ You have successfully launched a complaint

}
[Object: null prototype] {
  name: 'Saarthak Goyal',
  email: 'saarthak756.be22@chitkara.edu.in',
  contact: '+91 95014 68460',
  desc: "I have lost my laptop's charger and i have doubt someone stole it ."
}
```

7. CONCLUSION

The Smart Complaint Management System successfully provides a modern and efficient solution for handling complaints and managing administrative tasks digitally. The system reduces manual workload, improves transparency, and enables faster issue resolution. Residents can easily register complaints and track their status, while administrators can manage operations efficiently from a centralized platform.

The project demonstrates the effective use of modern web technologies in solving real-world management challenges. By integrating complaint tracking, announcements, staff management, and salary handling into one platform, the system improves communication and operational efficiency significantly.

8. FUTURE SCOPE

The Smart Complaint Management System can be further enhanced with advanced features and technologies in the future. Planned improvements include:

- **Mobile Application:**

A dedicated Android and iOS application can improve accessibility and user engagement.

- **AI-Based Complaint Prioritization:**

Artificial Intelligence can be used to prioritize complaints based on urgency and category.

- **Push Notifications:**

Real-time notifications can be integrated using email and SMS services.

- **Cloud Deployment:**

Cloud hosting can improve scalability, performance, and availability.

- **Analytics Dashboard:**

Advanced charts and reports can help management analyze complaint trends and service quality.

- **Multi-Language Support:**

The system can support multiple languages for wider accessibility.

9. REFERENCES

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- JavaScript Documentation – <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
- Node.js – <https://nodejs.org/>
- Express.js – <https://expressjs.com/>
- MySQL Documentation – <https://www.mysql.com/>
- GitHub – <https://github.com/>